Introduction

The sampling distribution of a statistic is the distribution of its values across hypothetical repetitions of the sampling process. Every confidence interval, every p-value, and every standard error is a statement about the sampling distribution of some statistic. Understanding this concept makes inference intuitive; ignoring it makes it a ritual of incantations.

Prerequisites

The reader should know what a sample mean is and should be comfortable with histograms and R's `replicate()` function.

Theory

Let $T = T(X_1, \ldots, X_n)$ be a statistic computed from a sample of size n drawn from a population with some distribution. The **sampling distribution** of T is the distribution of values T would take if the sampling process were repeated many times, each time drawing a fresh sample of size n from the same population.

Key facts:

- The sampling distribution is a property of T and of the sampling design, not of any particular observed sample.
- Its mean is the expectation $E[T]$. For unbiased estimators, $E[T]$ equals the population parameter.
- Its standard deviation is called the **standard error** of T: $\mathrm{SE}(T) = \sqrt{\mathrm{Var}(T)}$.
- Its shape depends on the statistic, on n, and on the population distribution. For the sample mean from a population with finite variance, the central limit theorem guarantees approximate normality as n grows.

For the sample mean $\bar{X}$ of n iid observations with mean μ and variance σ^2:

$$E[\bar{X}] = \mu, \qquad \mathrm{SE}(\bar{X}) = \sigma/\sqrt{n}.$$

The standard error tells us the typical distance between $\bar{X}$ and μ – it is the unit in which confidence intervals are measured.

Assumptions

The theoretical sampling distribution derived in most textbooks assumes iid sampling from a population with finite variance. Deviations (heavy tails, dependence, non-iid sampling) change the shape or the formulas; the bootstrap lets us approximate the sampling distribution empirically without these assumptions.

R Implementation

```r
library(ggplot2)

set.seed(2026)
population <- rgamma(1e5, shape = 2, rate = 0.1)
mu <- mean(population)
sigma <- sd(population)

draw_mean <- function(n) mean(sample(population, n))

reps <- 10000
means_n10  <- replicate(reps, draw_mean(10))
means_n100 <- replicate(reps, draw_mean(100))
```

```r
cat(sprintf("n=10  : mean=%.2f, SE=%.2f; theoretical SE=%.2f\n",
            mean(means_n10), sd(means_n10), sigma / sqrt(10)))
cat(sprintf("n=100 : mean=%.2f, SE=%.2f; theoretical SE=%.2f\n",
            mean(means_n100), sd(means_n100), sigma / sqrt(100)))

df <- data.frame(xbar = c(means_n10, means_n100),
                 n = factor(rep(c(10, 100), each = reps)))
ggplot(df, aes(x = xbar, fill = n)) +
  geom_histogram(alpha = 0.6, bins = 50, position = "identity") +
  labs(x = expression(bar(X)), y = "Frequency") +
  theme_minimal()
```

For a skewed gamma population, the sampling distribution of the mean concentrates as n grows, and its spread matches the theoretical standard error.

Output & Results

```
n=10  : mean=19.98, SE=4.42; theoretical SE=4.47
n=100 : mean=20.01, SE=1.41; theoretical SE=1.41
```

Increasing n by a factor of 10 decreases the standard error by $\sqrt{10} \approx 3.16$, exactly as the $1/\sqrt{n}$ rule predicts.

Interpretation

The standard deviation of the sampling distribution – the SE – is what a confidence interval is built from. A 95% CI for the mean is approximately $\bar{x} \pm 1.96 \cdot \text{SE}(\bar{X})$. When a manuscript reports "mean 19.98 (SE 1.41, 95% CI 17.22–22.74)", every number in that summary is derived from the sampling distribution of the statistic.

Practical Tips

- Do not confuse the standard deviation of the data with the standard error of a statistic. The SD describes variability across individuals; the SE describes variability across hypothetical samples of size n.
- When the theoretical sampling distribution is unknown, approximate it by the bootstrap: resample with replacement from the observed sample, compute the statistic, and use the empirical distribution of the resampled statistics as an estimate.
- The sampling distribution depends on n – larger samples yield tighter distributions, as the $1/\sqrt{n}$ rule captures for the mean.
- For statistics other than the mean, the sampling distribution may be harder to derive analytically but is equally well-defined; simulation or bootstrap is the general-purpose tool.
- The t distribution that appears in small-sample inference is the sampling distribution of $(\bar{X} - \mu)/(s/\sqrt{n})$ under normality with unknown variance, not of the mean itself.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Scientific process and research workflow
- R, RStudio, Quarto, and renv toolchain
- Data types, tidy data, accuracy and precision
- Import, joins, and missingness with dplyr